

TECHNICAL ADVANCE

Whole-tissue 3D immunostaining of shoot apical meristems in rice at single-cell resolution

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SUMMARY

The shoot apical meristem (SAM) produces all above-ground organs of plants and is thus a central focus of plant developmental biology. Developmental processes in the SAM are regulated by various factors that control gene expression at the cellular level. Key among these are the chemical modifications of the N-terminal tails of histones, which are essential components of nucleosomes and chromatin that play crucial roles in these processes. While immunostaining is a valuable method for the spatial analysis of histone modifications, its application to the SAM has posed technical challenges. Here, we developed a three-dimensional immunostaining method for rice (*Oryza sativa*) SAMs at single-cell resolution using a permeabilization process with specific cell wall degrading enzymes, along with the iTOME1 clearing technique (Sakamoto et al. [2022] *Communications Biology*, 5, 12). We detected clear signals throughout the deeper tissue layers, allowing us to visualize histone modifications associated with both active and repressive chromatin states, as well as M phase-specific modifications localized on chromosomes. The repressive modifications H3K9me2 and H3K27me3 exhibited punctate patterns within the nuclei, whereas the modifications linked to transcriptional activity were more diffusely distributed. Double staining showed that H3K9me2 forms a peripheral layer around a central domain enriched in H3K4me1. A comparative analysis of SAMs during the vegetative and reproductive phases indicated that active modifications persisted across both phases, whereas repressive modifications increased during the reproductive phase. Our protocol facilitates the three-dimensional visualization of chromatin states in the SAM, offering a robust tool for exploring the spatial regulation of plant development at the single-cell level.

Keywords: epigenetics, flowering, imaging, immunostaining, rice, shoot apical meristem.

INTRODUCTION

All above-ground organs in plants originate from the shoot apical meristem (SAM), a stem cell tissue with a dome-like structure (Fuchs & Lohmann, 2020; Tanaka et al., 2023). The characteristics of organ formation from the SAM change during developmental phase transitions (Tsuji & Sato, 2024). During the vegetative phase, the SAM produces leaves. After transitioning to the reproductive phase, the SAM shifts to forming reproductive organs that develop into flowers. This dynamic transition helps define organ identity and the spatial-temporal arrangement of organs during development (Tsuji & Sato, 2024). The three-dimensional

(3D) distribution of some plant hormones and transcription factors involved in hormone signaling within the SAM has been established (Demesa-Arevalo et al., 2024). Local auxin maxima predict the sites of leaf primordium initiation (Galavotti et al., 2008; Reinhardt et al., 2003; Yang, Yuan, et al., 2017), and cytokinin accumulation at the tip of the SAM induces *WUSCHEL* gene expression, which helps maintain stem cell activity (Gordon et al., 2009). Lateral organ differentiation in the SAM occurs in a 3D pattern specific to the developmental phase of the SAM, with the underlying molecules positioned appropriately within the SAM in three dimensions.

Epigenetic modifications constitute an additional crucial layer in the regulation of SAM development (Nguyen & Gutzat, 2022). These modifications ensure that the gene expression patterns required for proper development are established, playing key roles in creating euchromatic and heterochromatic domains, as well as in silencing transposable elements (Hemenway & Gehring, 2023). Typical epigenetic modifications include DNA methylation and histone modifications; the latter are particularly associated with dynamic changes in gene expression (Liu et al., 2015; You et al., 2017). Histone modifications undergo notable changes in the SAM. For example, two distinct epigenetic states have been identified in *Arabidopsis* (*Arabidopsis thaliana*) SAMs (You et al., 2017). The “E-state” is defined by narrow regions of histone H3 lysine 27 trimethylation (H3K27me3) embedded within broad H3K4me3 domains and is specific to the SAM. By contrast, the “H-state” features broad H3K27me3 regions containing narrow H3K4me3 regions; this configuration undergoes dynamic changes during the vegetative to reproductive transition. In rice (*Oryza sativa*) SAMs, the H3K27me3-to-H3K4me3 ratio is thought to regulate gene expression during the developmental phase (Liu et al., 2015). However, despite the accumulation of sequence-based information from bulk samples, the 3D distribution of epigenetic modifications within the SAM remains unknown.

Several methods are available for observing the 3D distribution of histone modifications, including the use of reporter lines and immunostaining (Araki et al., 2022; Kurita et al., 2017; Nagaki et al., 2017). Reporter lines utilizing the modification-specific intracellular antibody (mintbody) method (Kurita et al., 2017) enable live-cell imaging of dynamic changes in specific histone modifications. However, their application depends on the availability of transformation techniques, the integrity of the chosen promoters, and the successful expression of functional proteins in plants. Another approach, immunostaining, is suitable for plants that cannot be genetically transformed and allows for the precise localization of endogenous targets (Araki et al., 2022; Nagaki et al., 2017). Moreover, immunostaining visualizes the distribution of molecules and modifications for which reporter systems are not yet available. However, this method requires reliable antibodies and efficient permeabilization procedures, which must be optimized for each plant species and tissue type. These challenges have hindered the development of immunostaining protocols for several important tissues, including the SAM.

The rice SAM is located in the basal region of the shoot, positioned at the tip of the unelongated stem during the vegetative phase (Figure 1a,b). The SAM is composed of several hundred cells that form a dome-like structure with a diameter of approximately 50 μm during the vegetative phase (Figure 1c–f). This structure expands after the

phase transition. Due to the high cell density of the rice SAM, performing 3D immunostaining on this tissue has been challenging.

Here, we introduce a method for 3D immunostaining of rice SAMs. Compared with previous staining procedures, this method includes several critical improvements that enhance the permeabilization efficiency and reduce background signals. Using this approach, we revealed the distribution of histone modifications in the SAM and demonstrated how these modifications change during the transition from the vegetative phase to the reproductive phase.

RESULTS

Development of 3D immunostaining for rice SAMs

In this section, we summarize our immunostaining method and the key steps we optimized to enable the visualization of fluorescence signals throughout the SAM (Table 1, see Data S1 for details). We examined SAMs during the vegetative and reproductive phases, as defined in Figure S1.

First, we sampled SAMs from rice plants by removing all mature leaves to expose the tissue (Figure 1h; Movies S1 and S2). Since the SAM is very small, we included residual basal tissue below the SAM (as indicated by the dotted line in Figure 1h) to facilitate the handling of small samples during preparation and analysis. We then placed approximately 8–10 samples into a 1.5-mL tube (Figure 1i,j).

Next, we fixed the SAMs with 0.25–1.0% (v/v) formaldehyde (Figure S2, depending on the histone modification to be examined). This was followed by cell wall digestion using 2.5% (w/v) Pectolyase Y-23 (Figure 1g) and permeabilization with 1.0% (v/v) Triton X-100. Both treatments were performed at relatively high concentrations compared with the previous protocol (Araki et al., 2022; Bernal et al., 2019; Nagaki et al., 2017; Tran et al., 2021), which proved essential for effective antibody penetration (Figure 1g, Figure S3). Antigen retrieval using Proteinase K caused nuclear shrinkage under our conditions (Figure S4). During incubation, the uncapped 1.5-mL tube was sealed with Parafilm, punctured to allow degassing, and laid horizontally during incubation to prevent the SAMs from sticking to one another (Figure 1i,j).

After blocking with 1% (w/v) bovine serum albumin (BSA), we transferred the SAMs to a plastic dish and placed them on a glass depression slide (Figure 1k–m). Under a stereomicroscope, we carefully trimmed the residual basal tissue from the SAMs, thereby improving antibody accessibility (Figure 1n–p, Figure S5). Next, we performed primary and secondary antibody reactions in a moisture chamber (Figure 1q). We selected antibodies that exhibited specific binding without generating non-specific signals in the rice SAM (Figure S6). Adding 500 mM NaCl to the antibody reactions reduced non-specific signals (Figure S7). We then

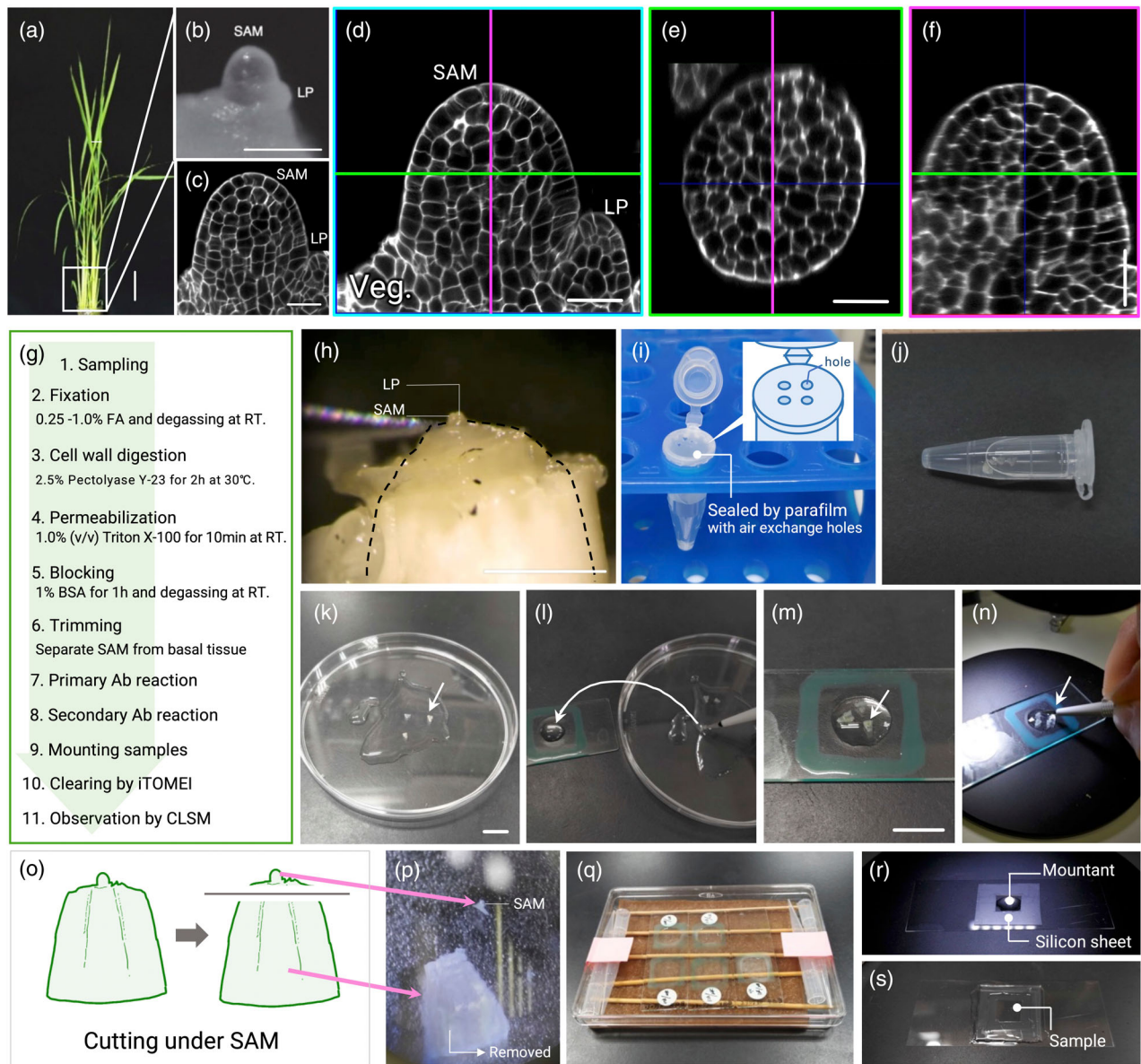


Figure 1. 3D imaging and overview of the 3D immunostaining protocol for the rice SAM.

(a, b) The rice shoot apical meristem (SAM) and leaf primordium (LP).

(c–f) A longitudinal section (c) and 3D observations (d–f) of the SAM using confocal laser scanning microscopy. Cell walls were counterstained with SCRI Renaissance Stain 2200 (SR2200, white). Orthogonal sections: XY (longitudinal section) (d), YZ (horizontal section) (e), and ZX (vertical section) (f). Veg., Vegetative phase.

(g) Overview of the 3D immunostaining protocol. CLSM, confocal laser scanning microscopy; FA, formaldehyde.

(h) Sampling of a SAM. Residual basal tissue below the SAM was included, as indicated by the dotted line.

(i) Microtube sealed with Parafilm containing air exchange holes for degassing.

(j) Microtubes laid horizontally to prevent the samples from sticking to one another during incubation.

(k–m) Transfer of samples to a glass depression slide using a scalpel via a plastic dish after blocking.

(n–p) Trimming of the basal tissue of the SAM using a scalpel under a stereomicroscope.

(q) Moisture chamber used during antibody incubation and clearing to prevent drying.

(r, s) Sample mounting for observation. A silicone sheet with a square hole was placed between two cover glasses. The samples and mountant were placed into the square hole. Scale bars, 10 cm (a), 100 μ m (b), 20 μ m (c–f), 500 μ m (h), and 1 cm (k, m).

cleared the tissues using the iTOIMEI method (Sakamoto et al., 2022; Sato et al., 2022) (Figure S8). To observe the SAMs, we prepared a slide using two cover glasses and a

silicone sheet and added a mountant to the square-shaped hole in the silicone sheet (Figure 1r). We transferred the SAMs into the mountant under a stereomicroscope, covered

Table 1 Comparison of 3D immunostaining methods

Step	Conditions used in this study	Conditions used in previous reports for maize SAM (Tran et al., 2021)	Conditions used in previous reports for rice anther (Araki et al., 2022)	Conditions used in previous reports for leaves of wheat, barley, maize, rice, and other species (Nagaki et al., 2017)	Related figures
Fixation	0.25–1.0% FA; degassed at 0.1 MPa for 2 min (×3) and 10 min (×2)	4% PFA	4% PFA; degassed at 0.09 MPa for 20 min on ice (×3 or more); incubated for 100 min at 25°C	3% PFA; degassed at 0.05 MPa for 2 min (×3); incubated for 15 min at RT	Figure 1; Figure S5
Cell wall digestion	2 h at 30°C	11 min at 25°C	1 min at 25°C	Degassed at 0.05 MPa for 2 min; incubated for 30–60 min at 37°C	Figure 1
Enzymes	2.5% pectolyase Y-23	1% Driselase, 0.5% cellulase, 1% pectolyase	2.0% cellulase Onozuka-RS, 0.3% pectolyase Y-23, 0.5% Macerzyme-R10	1% cellulase Onozuka RS, 0.5% pectolyase Y-23	
Permeabilization	1% Triton X-100 for 10 min at RT	2% Tween 20 for 2 h on ice	Not tested	2-propanol for 1 h at RT; ClearSee solution for 1–7 days at RT	Figure 1; Figure S2
Blocking	1% BSA; degassed 2 min (×5); incubated for 1 h at RT	3% BSA for 1 h at RT	3% BSA for 1 h at 25°C	Not described	Figure 1
Second trimming	Trimming at just beneath the SAM	Not tested	Not tested	Not tested	Figure 1; Figure S3
Primary Ab reaction	Degassed at 0.1 MPa for 2 min (×5) at RT; incubated for 16 h at 30°C	Incubated at least 2 h at 4°C	Degassed at 0.05 MPa for 2 min (×4 or more) at 25°C; incubated overnight at 4°C	Degassed at 0.05 MPa for 2 min (×5); incubated for 3–7 days at 4°C	Figure 1; Figures S4 and S7
Secondary Ab reaction	Degassed at 0.1 MPa for 2 min (×5) at RT; incubated for 3 h at RT	Incubated for 2 h to overnight at 4°C	Degassed at 0.05 MPa for 2 min (×4 or more) at 25°C; incubated overnight at 4°C	Degassed at 0.05 MPa for 2 min (×5); incubated for 3–7 days at 4°C	Figure 1
Mounting sample preparation	iTOMEI	Mounting medium (Prolong)	iTOMEI	An anti-fade, SlowFade Diamond containing 1 µg/mL DAPI (Thermo Fisher Scientific)	Figure 1
Clearing	iTOMEI	Not tested	iTOMEI	Integrated into permeabilization	Figure 1; Figure S8

Abbreviations: FA, formaldehyde; PFA, paraformaldehyde; RT, room temperature; BSA, bovine serum albumin.

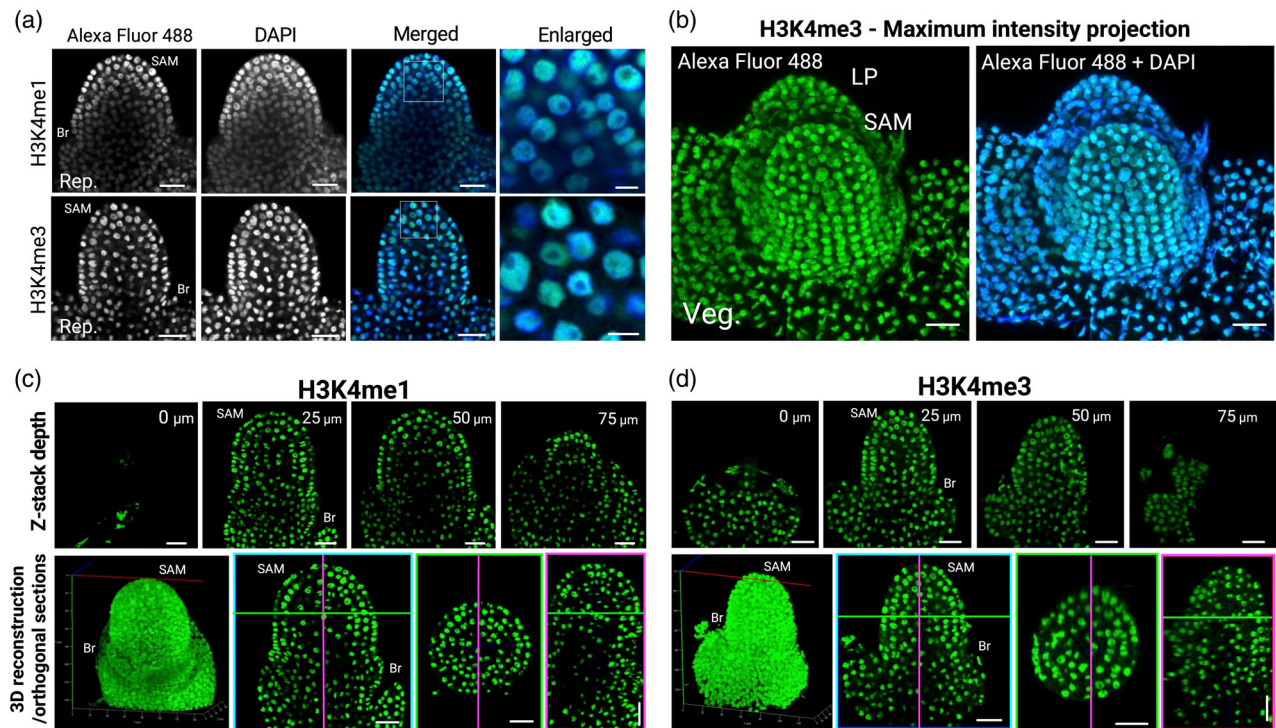


Figure 2. Spatial distribution of H3K4me1 and H3K4me3 in the rice SAM.

(a) Longitudinal sections of reproductive rice SAMs subjected to 3D immunostaining for H3K4me1 and H3K4me3. In the merged images, Alexa Fluor 488 (green) indicates histone modification signals, and DAPI (blue) indicates stained nuclei. Br, Bract. (b) Maximum intensity projection image of H3K4me3 immunostaining. Left: Alexa Fluor 488 (green). Right: Merged image with DAPI (blue). LP, leaf primordium. (c, d) Z-stack images and 3D reconstruction of the SAM stained for H3K4me1 (c) and H3K4me3 (d). Upper panels: Longitudinal sections of the SAM. Confocal optical sections were captured from the surface (0 μm) to the deepest layers of each sample. The signals were clearly recognized in the deeper sections. Lower panels: 3D reconstruction and orthogonal sections (XY, YZ, and ZX) derived from z-stack series. Orthogonal sections: XY (longitudinal section, indicated in blue), YZ (horizontal section, indicated in green), and ZX (vertical section, indicated in magenta). Scale bars, 20 μm (Alexa Fluor 488, DAPI, merged in panels (a), (b-d)) and 5 μm (enlarged in panel (a)).

them with a small cover glass, and sealed the edges of the cover glass with nail polish (Figure 1s). Finally, we observed the SAMs under a confocal laser scanning microscope. Notably, this protocol preserved immunostaining signals for over 140 days (Figure S8b, Data S1).

Comparing signal patterns among histone modifications

Next, we examined key histone modifications using our newly developed 3D immunostaining protocol. These included marks associated with active chromatin (H3K4me1 and H3K4me3) (Inagaki et al., 2017; Zhang et al., 2009; Zhao et al., 2020), repressive chromatin (H3K27me3 and H3K9me2) (Deal & Henikoff, 2010; Naumann et al., 2005; Zhao et al., 2020), and the M phase of the cell cycle (H3 serine 10 phosphorylation, H3S10p) (Liu & Nonomura, 2016).

Histone modifications associated with active chromatin states

H3K4me1 and H3K4me3 are known to be associated with transcriptionally active chromatin in plants (Inagaki et al., 2017; Zhang et al., 2009; Zhao et al., 2020). H3K4me1 is

typically enriched in gene bodies, while H3K4me3 accumulates around the transcription start sites of actively expressed genes. To visualize the distribution of these modifications in the rice SAM, we performed 3D immunostaining (Figure 2; Movies S3 and S4). Longitudinal sections through the central region of the SAM displayed uniform signals for both H3K4me1 and H3K4me3 across all cell layers (Figure 2a).

To determine whether this uniformity extended to layer 1 (L1) of the leaf primordia that differentiated from the SAM, we generated a z-stack projection image (Figure 2b). Note that the clearing steps were omitted in this experiment to better visualize L1. The resulting image showed consistent H3K4me3 signal intensity throughout L1 (Figure 2b). Additionally, the leaf primordia and surrounding tissues adjacent to the SAM displayed signal intensities comparable to those observed within the SAM itself (Figure 2b).

Further analysis of the z-stack series revealed that all cells throughout the entire depth displayed H3K4me1 and H3K4me3 signals (Figure 2c,d; Movies S3 and S4). We

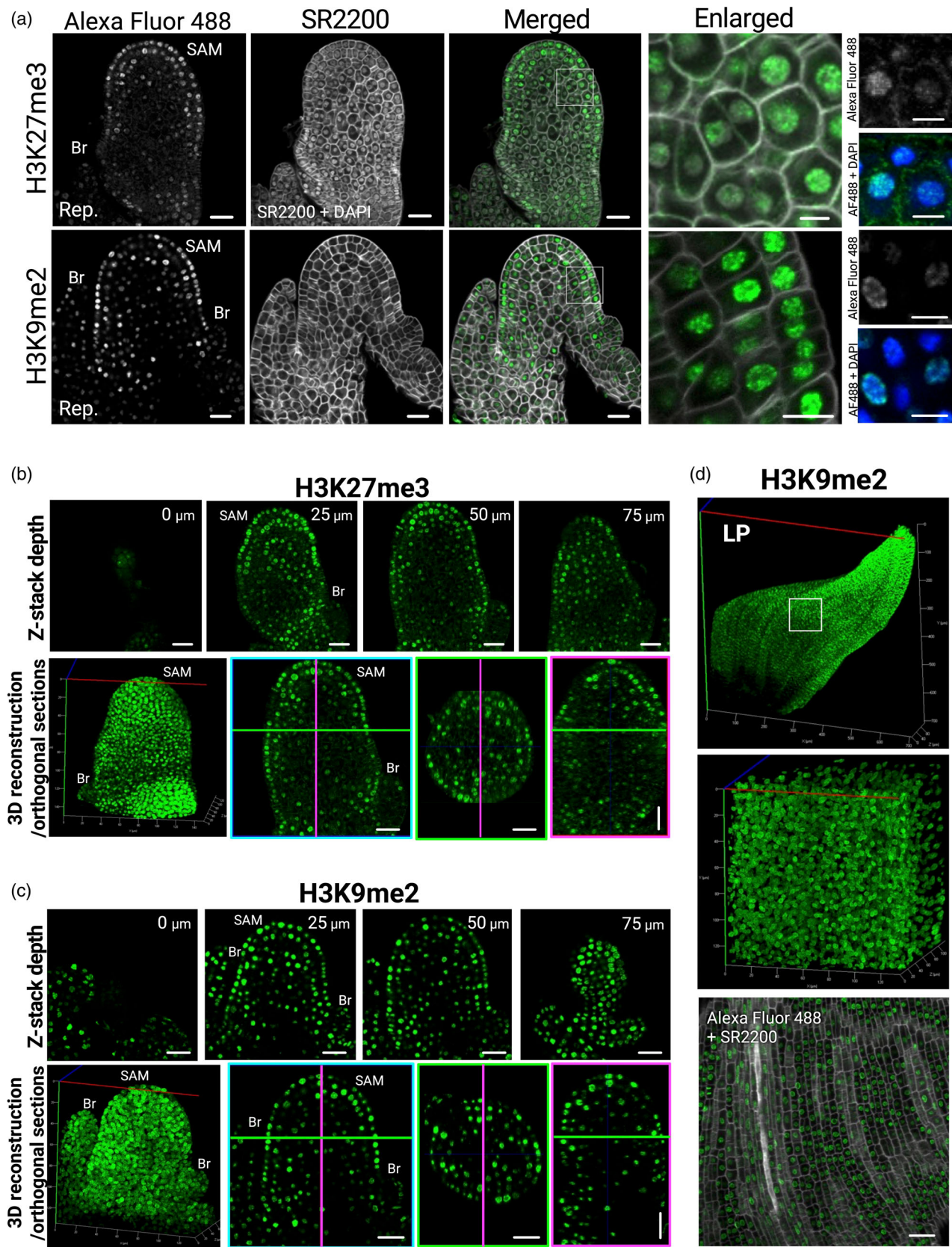


Figure 3. Spatial distribution of H3K27me3 and H3K9me2 in the rice SAM and leaf primordium.

(a) Longitudinal sections of reproductive rice SAMs subjected to 3D immunostaining for H3K27me3 and H3K9me2. In the merged images, Alexa Fluor 488 (green) indicates histone modification signals. SR2200 and DAPI (gray) were used to stain the cell walls and nuclei, respectively. Br, Bract. Far right: Differences in histone modifications across nuclei. Alexa Fluor 488 (green) and DAPI (blue) staining are shown in the merged images. Rep., Reproductive phase. (b, c) Z-stack images and 3D reconstruction of a SAM stained for H3K27me3 (b) and H3K9me2 (c). Upper panels: Longitudinal sections of the SAM. Confocal optical sections were obtained from the surface (0 μ m) to the deepest sections of each sample. The signals were clearly recognized in the deeper sections. Lower panels: 3D reconstruction and orthogonal sections (XY, YZ, and ZX) derived from the z-stack series. Orthogonal sections: XY (longitudinal section, indicated in blue), YZ (horizontal section, indicated in green), and ZX (vertical section, indicated in magenta). (d) 3D reconstruction and longitudinal sections of the rice leaf primordium. Top and middle: 3D reconstruction of the leaf primordium (top) and close-up view of the indicated area in the top panel (middle), stained for H3K9me2. Bottom: Longitudinal section of the LP showing H3K9me2 (green) and the cell wall (gray). LP, leaf primordium. Scale bars, 20 μ m (Alexa Fluor 488, DAPI, merged in panels (a), (b–d)) and 5 μ m (enlarged in the far right panels of (a)).

reconstructed 3D images from the z-stack data and confirmed the presence of H3K4me1 and H3K4me3 signals throughout the cells in both longitudinal and cross-sectional views (Figure 2c,d). These results suggest that our protocol allows for comprehensive 3D visualization of immunostaining signals, including those in the deeper layers of the rice SAM. Moreover, based on immunostaining, the distribution and intensity of active euchromatin were comparable across all cells comprising the SAM.

Histone modifications associated with repressive chromatin states

H3K27me3 and H3K9me2 are known to be associated with repressive chromatin (Deal & Henikoff, 2010; Naumann et al., 2005; Zhao et al., 2020). H3K27me3 is enriched in transcriptionally repressed genes and facultative heterochromatin, while H3K9me2 typically accumulates in transposable elements and constitutive heterochromatin.

H3K27me3 signals were detected within the nuclei of SAM cells (Figure 3a). A closer examination revealed variation in signal intensity among individual nuclei, suggesting heterogeneity in the distribution of facultative heterochromatin within the SAM (Figure 3a).

To determine whether this heterogeneity extended throughout the entire SAM, we acquired z-stack images and generated a 3D reconstruction (Figure 3b; Movie S5). The H3K27me3 signals appeared uneven across the z-stack series and in orthogonal sections derived from the 3D reconstruction, confirming that this heterogeneity was maintained throughout the SAM.

Next, we observed H3K9me2 signals in the rice SAM (Figure 3a). Upon closer examination of the stained cells, the signals appeared as dot-like structures concentrated in the perinuclear regions, consistent with the expected localization of H3K9me2 (Figure 3a) (Liu & Nonomura, 2016). We observed a distinctive pattern in which only discrete cells within the SAM exhibited H3K9me2 signals in their nuclei (Figure 3a). This pattern contrasts with observations for SAMs in other plant species and in fully developed organs, where H3K9me2 signals are typically detected uniformly across all cells within the tissue (Nagaki et al., 2017;

Takatsuka & Umeda, 2021). Analysis of z-stack images confirmed that this cell-to-cell heterogeneity persisted across all optical sections (Figure 3c; Movie S6).

To examine whether developing leaf primordia also exhibit similar spotty H3K9me2 signals, we observed the leaf primordia in rice samples (Figure 3d; Movie S7). The flat cellular arrangement in the leaf primordium aligned the nuclei at a similar focal depth, allowing us to directly compare signal intensities within a single optical section. In contrast to the SAM, H3K9me2 signals in the leaf primordium were more uniformly distributed (Figure 3d), indicating that the rice SAM displays a unique spatial distribution pattern of H3K9me2 compared with the developing leaf primordium.

In summary, the H3K27me3 and H3K9me2 signal intensities varied in each nucleus, suggesting that repressive histone modifications are not uniformly distributed in the rice SAM.

Validation of the spotty pattern in H3K9me2

H3K9me2 signals are typically uniformly distributed in plant tissues (Nagaki et al., 2017; Takatsuka & Umeda, 2021), which contrasts with our observation in the rice SAM (Figure 3a,c). We performed control experiments to determine whether the spotty H3K9me2 signal pattern resulted from technical artifacts.

First, to assess whether the spotty pattern was antibody-dependent, we compared the staining patterns using two different anti-H3K9me2 antibodies: ab1220 (Abcam) and MABI0317 (Fujifilm). Both antibodies produced similar spotty signal patterns (Figure 4a), suggesting that the observed distribution pattern was not antibody-dependent.

Next, we considered whether the spotty pattern might have been a misinterpretation caused by optical sectioning artifacts, such as partial detection due to the asymmetric localization of H3K9me2 within the nucleus. To test this possibility, we closely examined individual nuclei by extracting three optical sections at different z depths from the same z-stack and compared their signal intensities (Figure 4b). For H3K4me3, the signal intensities were consistent across all three sections in each nucleus (Figure 4c).

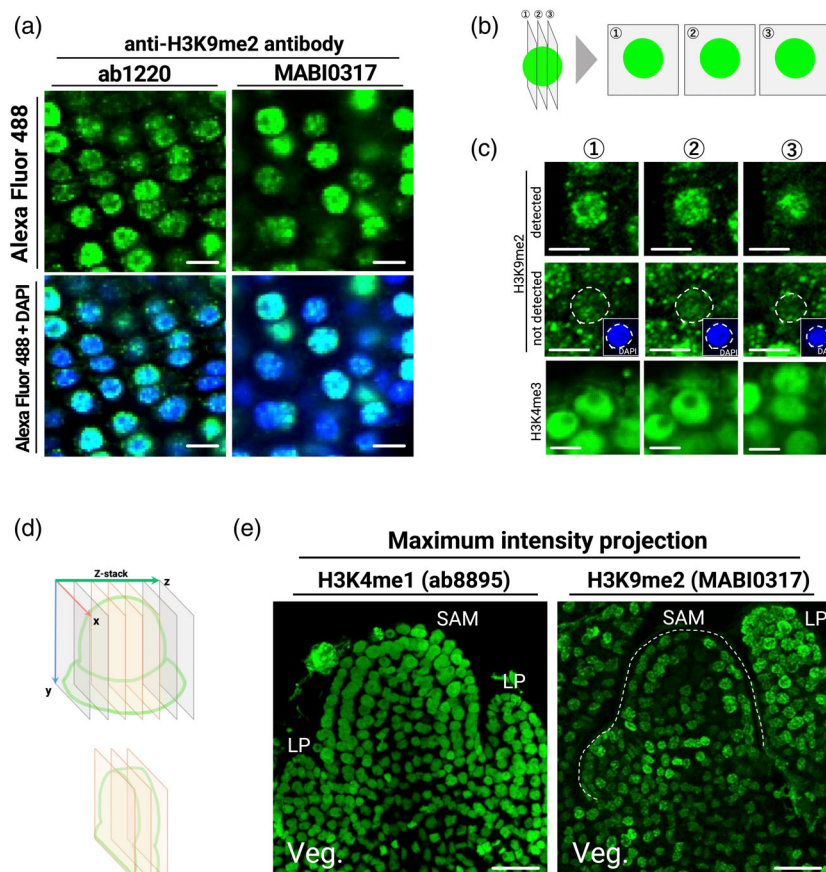


Figure 4. Validation of the spotty pattern observed by H3K9me2 immunostaining.

(a) Analysis of whether the spotty pattern is antibody-dependent. Merged images show H3K9me2 signals (Alexa Fluor 488, green) and nuclear staining (DAPI, blue).

(b) Diagram illustrating z-stack image acquisition across a single nucleus.

(c) Representative images from different z-depths (corresponding to the positions indicated in (b)) of a single nucleus stained for H3K9me2 and H3K4me3. Upper panels: A nucleus with detectable H3K9me2 signals. Middle panels: a nucleus without detectable H3K9me2 signals. Lower panels: a nucleus stained for H3K4me3, showing a consistent signal across depths.

(d) Diagram of z-stack acquisition in the entire SAM and a maximum intensity projection of the central 10-μm region.

(e) Maximum intensity projection images of the central 10-μm region from samples stained for H3K4me1 and H3K9me2. Scale bars, 5 μm (a, c) and 20 μm (e).

For H3K9me2, the nuclei were categorized into those with or without detectable signals. The presence or absence of the signal was consistent across all three z sections (Figure 4c). These results indicate that the spotty pattern was not due to depth-dependent partial detection of nuclei.

We further analyzed the maximum intensity projections of the z-stack series across the entire rice SAM, focusing on a central 10-μm stack along the apical-basal axis, which spanned an equivalent length of two nuclei (Figure 4d). In these projections, H3K4me1 signals were uniformly detected across all nuclei (Figure 4e). By contrast, H3K9me2 signals exhibited a spotty pattern (Figure 4e).

Collectively, these results indicate that the uneven distribution of H3K9me2 signals in the rice SAM is not attributable to antibody specificity, z-depth scanning artifacts, or

other technical issues but likely reflects biologically relevant heterogeneity (see Discussion).

Histone modifications associated with M phase

All modifications described above produced signals throughout the entire nucleus (Figures 2 and 3). To evaluate whether our protocol could also visualize localized signals inside the nucleus, we examined H3S10p, a modification known to accumulate in pericentromeric regions during M phase, where it typically appears as dot-like structures (Rocha et al., 2017). We conducted H3S10p immunostaining of the rice SAM. H3S10p signals were discretely distributed throughout the SAM (Figure 5a–d; Movie S8). Similar signal patterns were observed in the leaf primordium (Figure 5e). Upon closer inspection, H3S10p exhibited two distinct patterns: (1) localized

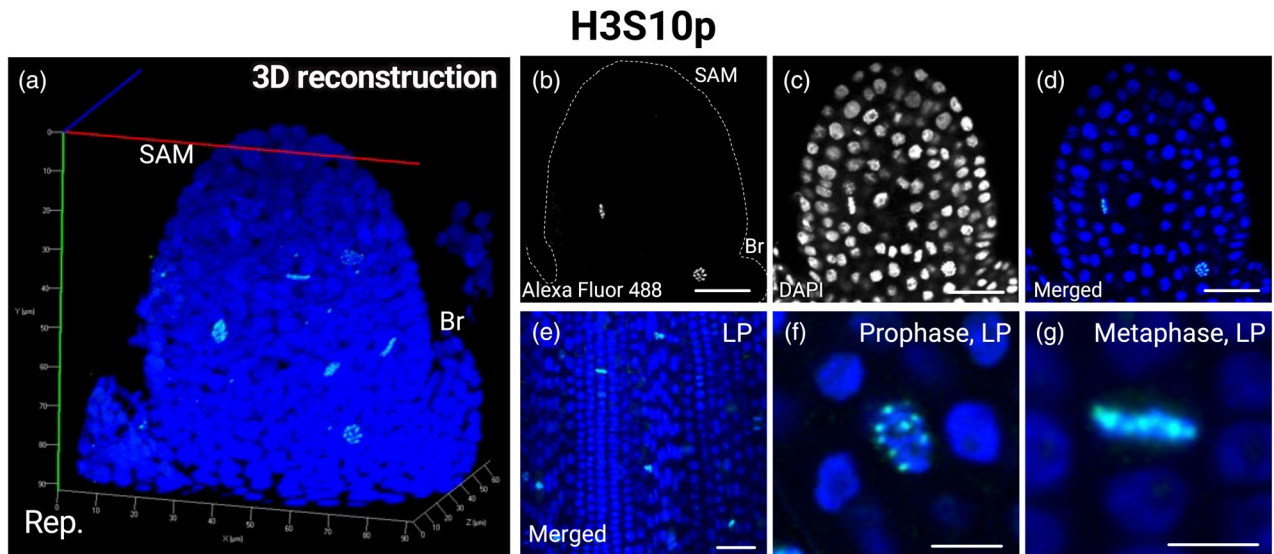


Figure 5. Spatial distribution of H3S10p in the rice SAM and leaf primordium.

(a) 3D reconstruction of H3S10p immunostaining in a rice SAM. Alexa Fluor 488 (green) indicates H3S10p signals, and nuclei were stained with DAPI (blue). Only M phase nuclei were labeled. (b-d) Longitudinal sections of 3D immunostaining showing H3S10p signals (b), DAPI staining (c), and a merged image of H3S10p (green) and DAPI (blue) (d). (e) Longitudinal sections of a leaf primordium stained for H3S10p with DAPI. LP, leaf primordium. (f, g) Localization of H3S10p signals in nuclei during prophase (f) and metaphase (g). Alexa Fluor 488 (green) indicates H3S10p, and the nuclei were stained with DAPI (blue). Scale bars, 20 μm (b-e) and 5 μm (f, g).

dot-like signals corresponding to pericentromeric regions during prophase (Figure 5f) and (2) aligned dot-like signals representing pericentromeric regions during metaphase (Figure 5g). These results indicate that our protocol is capable of visualizing localized signals within the nuclei.

Double staining of H3K4me1 and H3K9me2

Double staining is a useful approach for analyzing the spatial relationship between two histone modifications (Amar et al., 2024; Borg et al., 2020). To evaluate whether our protocol supports double staining, we performed simultaneous immunostaining of H3K4me1 and H3K9me2 in the rice SAM (Figure 6). Both signals were detected in the central regions of the SAM (Figure 6a). The signals were consistent with those observed in single-staining experiments: H3K4me1 signals appeared uniformly across all nuclei, while H3K9me2 signals were detected in a discrete subset of nuclei (Figures 2b, 3a-c and 6a,b). In merged images, this resulted in nuclei displaying only green fluorescence (H3K4me1) and nuclei showing both green and magenta fluorescence (H3K4me1 and perinuclear H3K9me2) (Figure 6a,b). These results confirm that our protocol enables effective double staining and provides reliable spatial resolution of distinct histone modifications in the rice SAM.

Double staining allowed us to examine the detailed spatial relationship between H3K4me1 and H3K9me2 within individual nuclei (Figure 6c,d). To determine

whether H3K9me2 is localized around H3K4me1, we measured the signal intensities of both modifications along a line drawn across the entire nucleus, including regions containing dot-like H3K9me2 signals (Figure 6c). As shown in Figure 6d, H3K9me2 exhibited peripheral enrichment outside the shoulders of the H3K4me1 intensity profile, indicating that H3K9me2 signals were distributed around H3K4me1. These results demonstrate that our protocol enables reliable double staining and allows for detailed spatial analysis of distinct histone modifications within the same nucleus.

Comparison of histone modifications in the SAM between the vegetative and reproductive phases

During the vegetative phase, the SAM continuously develops leaf primordia. Following the phase transition, the SAM shifts into the reproductive phase and begins inflorescence development (Tsuji & Sato, 2024). To investigate the differences in the spatial distribution and accumulation of histone modifications between the vegetative and reproductive phases, we compared immunostaining patterns in SAMs during both phases for each modification (Figure 7a).

The active modifications H3K4me1 and H3K4me3 exhibited strong and widespread signals during both phases (Figure 7a). By contrast, the repressive modifications H3K27me3 and H3K9me2 displayed increased signal intensity in the reproductive SAM compared with the

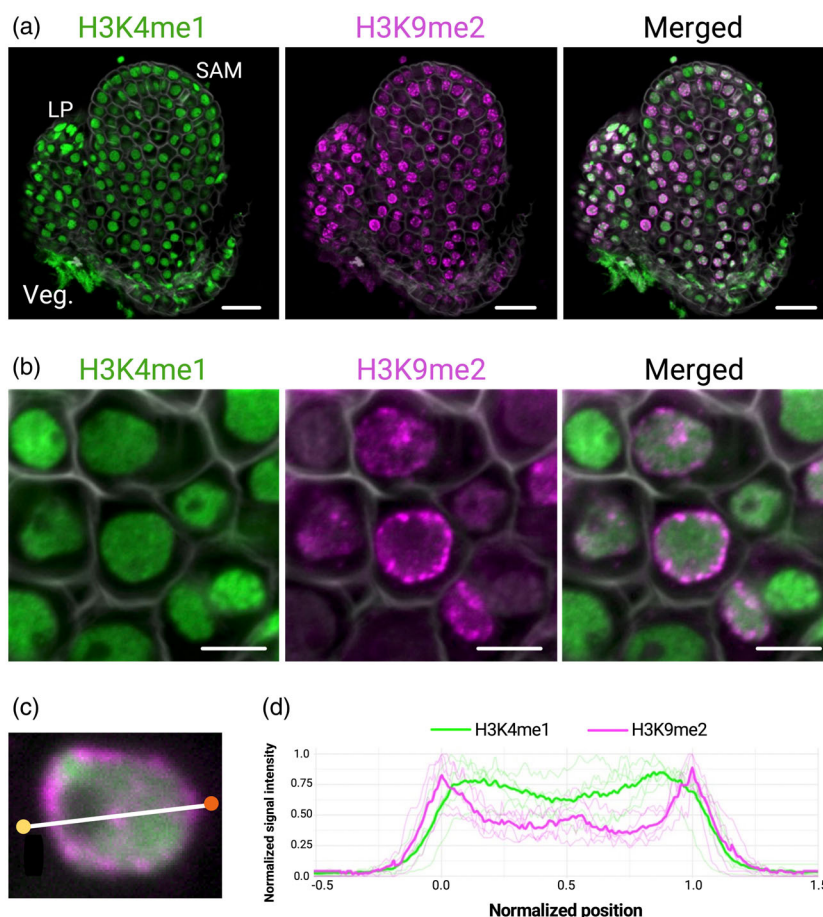


Figure 6. Double staining of H3K4me1 and H3K9me2 in the rice SAM.

(a) Longitudinal sections of a rice SAM subjected to 3D double immunostaining for H3K4me1 and H3K9me2. H3K4me1 (green), H3K9me2 (magenta), and the merged images are shown.

(b) Localization patterns of H3K4me1 and H3K9me2 in the nuclei.

(c, d) Quantification of H3K4me1 and H3K9me2 signals in the nuclei. An example of fluorescence measurement along a line connecting two perinuclear H3K9me2 signal peaks within a single nucleus (c). Profiles of normalized fluorescence intensities for H3K4me1 and H3K9me2 from five nuclei (thin lines) and the averaged profile (thick line) plotted against the normalized position between the two H3K9me2 peaks (d). Scale bars, 20 μ m (a) and 5 μ m (b).

vegetative SAM (Figure 7a). Additionally, H3K9me2 signals tended to be weak in the apical regions of the SAM during both phases. For H3S10p, signals were observed in M phase nuclei during both the vegetative and reproductive phases (Figure 7a). The frequency of H3S10p-positive nuclei was comparable between vegetative and reproductive SAMs during this early stage after the transition to the reproductive phase. These findings suggest that while active modifications are maintained, repressive modifications become more prominent during the vegetative to reproductive phase transition in the rice SAM.

Application of our 3D immunostaining method to other rice organs and meristems

To evaluate the applicability of our protocol to other rice tissues, we tested seven organs/meristems: the developing leaf sheath, mature leaf blade, axillary meristem,

developing spikelet, root epidermis, and root meristem (Figure S9). In the developing leaf sheath, H3K4me1 signals were detected in nearly all nuclei at depths exceeding 100 μ m (Figure S9a; Movie S9). In mature leaf blades, only a few cells showed signals in the epidermal tissue, whereas more cells exhibited signals in the regions adjacent to the edges formed during sample preparation (Figure S9b,c). By contrast, nuclei in the root epidermis and meristem were not detectable using our protocol (Figure S9d,e). Axillary meristems and their lateral organs at the P2 and P3 stages exhibited uniform H3K4me1 signals, with the prophyll showing stronger signals than the inner tissues (Figure S9f–h; Movie S10). In the developing spikelet, signals were detected in the palea and lemma, while the anther primordia and other tissues enclosed by the palea and lemma exhibited minimal or no signal (Figure S9i).

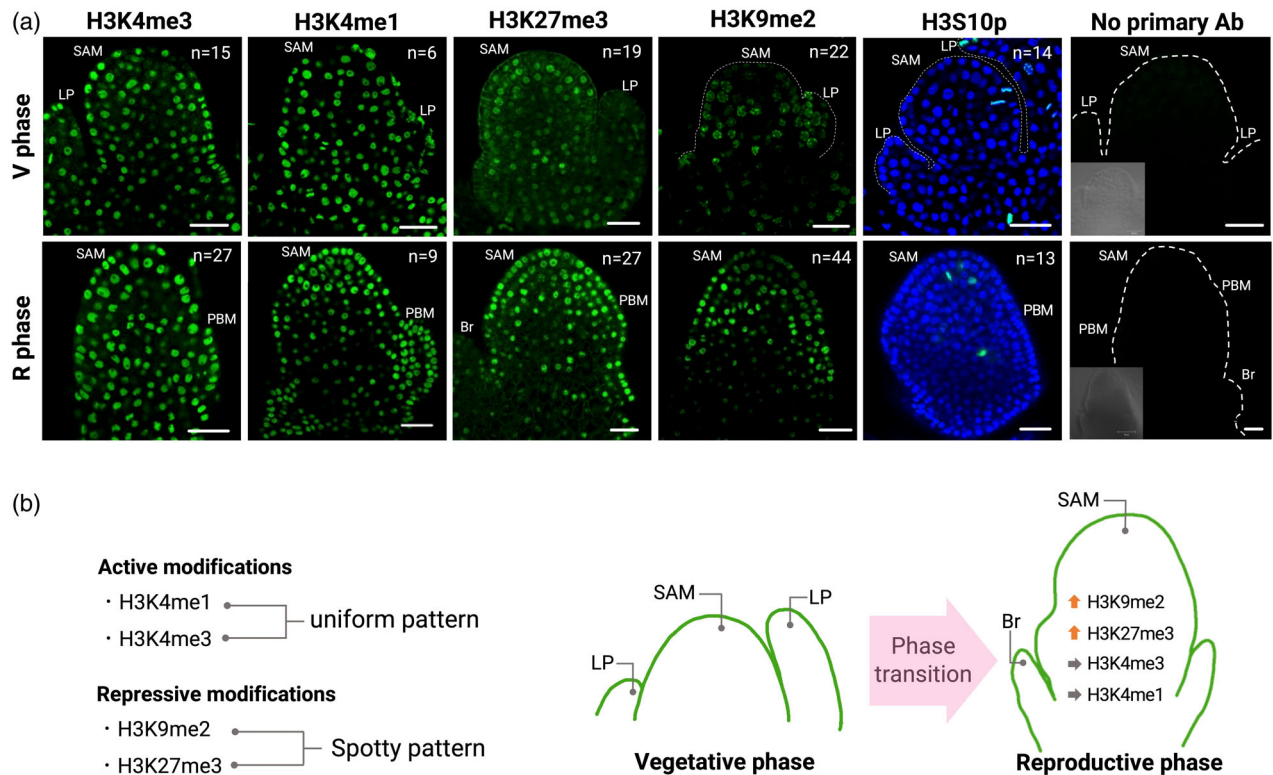


Figure 7. Comparison of histone modification patterns between vegetative and reproductive SAMs.

(a) Longitudinal sections of rice SAMs subjected to 3D immunostaining showing each histone modification during the vegetative (V; upper panels) and reproductive (R; lower panels) phases. H3S10p images include DAPI staining (blue).

(b) Diagram illustrating the patterns and changes in histone modifications in the SAM associated with the phase transition. Active modifications, such as H3K4me1 and H3K4me3, displayed a uniform pattern and accumulated similarly during both the vegetative and reproductive phases. By contrast, repressive modifications, such as H3K9me2 and H3K27me3, exhibited a spotty pattern with increased signal intensity following the transition to the reproductive phase. Scale bars, 20 μ m.

These results suggest that our protocol is broadly applicable to undifferentiated or actively dividing tissues, such as developing leaf sheaths, axillary meristems, and developing lemmas and paleae, but is less effective in highly differentiated tissues and roots.

DISCUSSION

The L1 layer functions as a barrier to antibody penetration inside the SAM

In this study, we developed a 3D immunostaining method for rice SAMs, allowing us to visualize the spatial distribution of histone modifications at single-cell resolution (Figures 1–3 and 5). We focused on the epidermal layer of the SAM (L1), which attenuates antibody penetration. Although the L1 cells were in direct contact with the antibody solutions, these solutions could not penetrate through the L1: immunostaining signals within the inner layers of the SAM remained weak unless the basal tissue was trimmed (Figure S5). This observation suggests that L1 acts as a barrier to antibody penetration. We propose

that this barrier effect is due to the presence of cuticular waxes, as treatment with 1.0% (v/v) Triton X-100, an effective detergent for solubilizing lipophilic components, enhances antibody penetration (Figure S3c,d). Consistent with this hypothesis, a gene encoding the enzyme fatty acid elongase, which is involved in the biosynthesis of very long-chain fatty acids in plant cuticular waxes, is specifically expressed in L1 of the rice SAM (Ito et al., 2011). These observations suggest that removing or disrupting extracellular waxes is important for enhancing antibody accessibility in plant tissues.

Heterogeneity of repressive histone modifications in the rice SAM

We successfully visualized H3K4me1 and H3K4me3 signals throughout the rice SAM, indicating that these active histone marks are uniformly distributed throughout this tissue (Figure 2). By contrast, the repressive modifications H3K27me3 and H3K9me2 exhibited heterogeneous signal intensities among cells (Figures 3 and 4). The underlying mechanism responsible for this heterogeneity remains

unclear. Although the cell cycle influences chromatin states, most known histone modifications are maintained throughout the cell cycle, except during the M phase, making the cell cycle an unlikely cause of the observed heterogeneity (Figure 5; Halsall et al., 2021; Hublitz et al., 2009). Interactions among histone modifications may also contribute to this heterogeneity; however, the previously reported antagonism between H3K9me2 and H3K4me1, as observed by ChIP-seq (Inagaki et al., 2017; Zhao et al., 2020), was only noted in the perinuclear regions in our immunostaining assays (Figure 6). Notably, both weaker fixation and Proteinase K treatment reduced the heterogeneity of repressive modification signals (Figures S2 and S4). These results suggest that unknown inhibitory factors that are sensitive to fixation are retained in specific cells and that Proteinase K might interfere with the access of antibodies against H3K9me2 and H3K27me3. Such factors could underlie the observed heterogeneity of repressive chromatin in the rice SAM. Further studies will be required to identify the molecular nature and mechanistic basis of this heterogeneity.

What is the biological significance of the observed heterogeneity in repressive histone modifications during rice SAM development? Cells with reduced H3K9me2 levels may exhibit increased expression of transposable elements (TEs), leading to the production of small interfering RNAs (siRNAs) from these elements. These siRNAs might act in a non-cell-autonomous manner to suppress TE expression in neighboring cells (Long et al., 2021). Indeed, substantial accumulation of TE-derived transcripts and elevated DNA methylation levels have been reported in the rice SAM (Higo et al., 2020; Tamaki et al., 2015), suggesting that TEs are actively regulated in this region. This phenomenon is reminiscent of the inter-tissue communication observed between tapetum cells and the male germline, in which siRNAs derived from TEs in the tapetum are transported to the male germline to reinforce TE silencing (Long et al., 2021). While the rice SAM represents intercellular communication within a single tissue, rather than inter-tissue signaling, the underlying principle—siRNA-mediated reinforcement of TE silencing—might be common to both situations.

Repressive modifications increase after the reproductive phase transition

By comparing signals between SAMs in the vegetative and reproductive phases, we observed enhanced repressive modifications, such as H3K27me3 and H3K9me2, following the transition from the vegetative to reproductive phases (Figure 7a,b). The increase in repressive modifications after the floral transition is consistent with the rise in CHH methylation levels (Higo et al., 2020). This suggests that the repressive context is stronger after the phase transition. The heightened levels of repressive marks, particularly

H3K27me3, may contribute to the suppression of transposons during the reproductive phase (Figure 7a; Déléris et al., 2021; Hisanaga et al., 2023). Indeed, several hundred transposons are repressed following the vegetative to reproductive transition in the rice SAM (Tamaki et al., 2015). While CHH methylation affects approximately half of these transposons, the other half does not exhibit changes in DNA methylation (Higo et al., 2020). Therefore, DNA methylation alone may not be the primary mechanism driving the suppression of these transposons. By contrast, both H3K27me3 and H3K9me2 are associated with the repression of transposons (Hisanaga et al., 2023; Déléris et al., 2021). This suggests that the observed increases in H3K27me3 and H3K9me2 may play a crucial role in transposon silencing during the reproductive phase.

The increase in repressive modifications may also contribute to the irreversibility of the vegetative to reproductive phase transition. In general, this transition is a one-way process, as it is estimated that the SAM does not revert to the vegetative phase (Adrian et al., 2009; Yang, Berry, et al., 2017). The process of vernalization in *Arabidopsis* is a well-known example of an irreversible developmental transition mediated by repressive modifications (Yang, Berry, et al., 2017). Before vernalization, *FLC* is highly expressed. However, after sufficient exposure to low temperatures, H3K27me3 accumulates at the *FLC* locus, leading to its irreversible repression (Yang, Berry, et al., 2017). Therefore, H3K27me3 might be associated with the irreversibility of the vegetative to reproductive transition in the rice SAM. These findings underscore the effectiveness of our approach in investigating the spatial regulation of plant development at single-cell resolution.

Potential applications of our protocol for studying SAM identity and the phase transition in the SAM

Our protocol can provide insights into three aspects of the developmental biology of the SAM: understanding cellular heterogeneity, elucidating the spatial distribution of histone modifications and their readers, and visualizing intrinsic signals without the need for available reporters. First, heterogeneity in epigenetic marks can be observed at the whole-nucleus level (Figures 3, 4 and 7). Such heterogeneity is difficult to capture using single-cell isolation and sequencing because these approaches lose the spatial context of epigenetic heterogeneity. By performing 3D immunostaining of the SAM, we can investigate the roles of epigenetic heterogeneity associated with the phase transition. Second, this method can reveal the spatial localization of histone modifications and associated chromatin regulators through simultaneous immunostaining. Such an analysis could help clarify the interplay between epigenetic modifications and their readers during the reproductive phase transition. Third, this 3D immunostaining protocol provides spatial visualization of signals in the

absence of an available transformation-based reporter system. This includes visualization of the phosphorylation of regulatory proteins, such as the phosphorylation of serine residues in RNA polymerase II, which is a marker of active transcription (Hajheidari et al., 2013). Combined with such techniques, our 3D immunostaining method could help elucidate SAM identity based on the epigenetic states of individual cells in the future.

MATERIALS AND METHODS

Plant materials and growth conditions

The *japonica* rice (*Oryza sativa*) cultivar Nipponbare was used in this study. All plants were grown in climate chambers with 70% relative humidity under short-day conditions consisting of daily cycles of 10 h of light at 28°C and 14 h of darkness at 25°C. Light was provided by fluorescent white light tubes (400–700 nm, 100 $\mu\text{mol m}^{-2} \text{s}^{-1}$). Plants sampled between 14 and 17 days and 26 and 30 days after sowing were used to collect vegetative and reproductive SAMs, respectively (Figure S1).

Immunostaining

Sampling and fixation

To collect shoot apices, several leaves were carefully removed from the basal parts of the plants through manual sectioning under a stereomicroscope to expose the P1 leaf primordium during the vegetative phase and the SAM during the reproductive phase. The samples were excised and fixed in a microtube containing 0.25% (v/v) formaldehyde or the indicated concentrations (Figure S2) in 1 mL of PEMT buffer (50 mM PIPES, 2 mM EGTA, 2 mM $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 0.05% [v/v] Triton X-100; pH 7.2). Degassing was performed under reduced pressure (0.08–0.1 MPa) at room temperature, consisting of three 2-min cycles and two 10-min cycles. During degassing, the microtubes were sealed with Parafilm that contained air exchange holes (Figure 1i). After fixation, the samples were washed three times with 1 mL of PEMT buffer for 5 min each at room temperature. From this step onward, the microtubes were laid horizontally to prevent the samples from sticking to one another during incubation (Figure 1j).

Cell wall digestion and permeabilization

To digest the cell wall, fixed samples were incubated in 2.5% (w/v) Pectolyase Y-23 (Seishin Pharmaceutical Co., Ltd., currently available from Kyowa Chemical, Osaka, Japan) in 1 mL of PEMT buffer containing 0.4 M mannitol for 2 h at 30°C. Following digestion, the samples were washed three times with 1 mL of PEMT buffer for 5 min each at room temperature. To enhance antibody permeabilization, the samples were then incubated in 1.0% (w/v) Triton X-100 in 1 mL of PEM buffer (50 mM PIPES, 2 mM EGTA, 2 mM $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$; pH 7.2) for 10 min at room temperature. After permeabilization, the samples were washed three times with 1 mL of glycine buffer (130 mM NaCl, 5.1 mM $\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$, 1.6 mM KH_2PO_4 , 50 mM glycine; pH 7.4) for 5 min each at room temperature.

Blocking, trimming, and antibody treatment

After permeabilization, the samples were blocked in 1 mL of 1% (w/v) BSA in glycine buffer. Degassing was performed five times for 2 min each, followed by incubation for 30 min at room temperature.

After blocking, the samples were transferred to a plastic dish for easier handling, gently picked up with a scalpel, and placed onto a glass depression slide (Figure 1k–m). During this step, the residual basal tissue was carefully trimmed away in 1% (w/v) BSA buffer under a stereomicroscope (Figure 1n–p). From this point onward, all buffer exchanges were performed under a stereomicroscope to avoid sample loss.

For the primary antibody reaction, the buffer was replaced with a primary antibody solution (the primary antibody was diluted 1:100 in 100 μL of 1% [w/v] BSA in glycine buffer; the primary antibodies used in this study are listed in Table S1). To suppress non-specific fluorescence, 500 mM NaCl was included in the primary antibody solution when necessary. The samples were degassed five times for 2 min each and incubated for 16 h at 30°C in a moisture chamber. Following removal of the primary antibody, the samples were washed three times with 100 μL of glycine buffer for 5 min each. For the secondary antibody reaction, the buffer was replaced with the secondary antibody solution (the secondary antibody was diluted 1:300 in 100 μL of 1% [w/v] BSA in glycine buffer; the secondary antibodies used in this study are listed in Table S1). Degassing was performed five times for 2 min each in a moisture chamber wrapped in aluminum foil to avoid light exposure. The samples were then incubated for 3 h at 30°C in a dark moisture chamber.

Sequential double immunostaining for H3K4me1 and H3K9me2

Sequential double immunostaining was performed to detect H3K4me1 and H3K9me2. Antibody incubation steps were performed as described in the “Blocking, trimming, and antibody treatment” section, with the following modifications.

After blocking and trimming, the first primary antibody reaction was conducted using anti-H3K9me2. Following antibody removal and three washes with glycine buffer, the second primary antibody reaction was performed using anti-H3K4me1 for 3 h. After removing the antibody and performing three additional washes, the secondary antibody reactions were conducted sequentially. First, the secondary antibody reaction for H3K9me2 was conducted. Following antibody removal and three washes, the secondary antibody reaction for H3K4me1 was performed.

Clearing

After conducting the antibody reactions, the samples were cleared using the iTOME1 method (Sakamoto et al., 2022). The samples were placed in 100 μL of destaining solution (100 mM sodium phosphate buffer, pH 8.0, with 20% [w/v] caprylyl sulfobetaine [TCI, Tokyo, Japan]) containing 2 $\mu\text{g/mL}$ of 4',6-diamidino-2-phenylindole (DAPI; Roche, Basel, Switzerland) and 0.1% (v/v) SCRI Renaissance Stain 2200 (SR2200, cell wall stain; the concentration of the solution from the supplier was considered to be 100%; Renaissance Chemicals, North Yorkshire, UK) to visualize nuclei and cell walls. The samples were incubated for 24 h at room temperature in a dark moisture chamber.

For imaging, a mounting setup was prepared using two cover glasses and a silicone sheet with a square hole. The silicone sheet was placed on a glass slide, and destaining solution was added to the square hole. The mounted samples were transferred to destaining solution in this setup, and the solution was exchanged with clearing solution (70.4% (w/w) Histodenz [Merck, Darmstadt, Germany] in PBS) as a mountant under a stereomicroscope. This step was critical for locating the samples, which became completely transparent upon exposure to clearing

solution. Finally, the samples were incubated for 5 min at room temperature in a dark moisture chamber and covered with another glass slide for observation.

Confocal laser scanning microscopy and imaging

The samples were visualized under a confocal laser scanning microscope (LSM700 and LSM880; Carl Zeiss, Oberkochen, Germany; TCS SP8; Leica Microsystems, Tokyo, Japan) equipped with 405-, 488-, and 555-nm laser lines and a 40 $\times$ 1.30 N.A. oil immersion objective (LSM700); 488- and 543-nm laser lines, a 20 $\times$ 0.8 N.A. dry objective, and a 63 $\times$ 1.40 N.A. oil immersion objective (LSM880); and a 405-nm laser line and a pulsed white light laser (WLL) source and a 63 $\times$ 1.30 N.A. glycerol-immersion objective lens (TCS SP8). For DAPI and SR2200 fluorescence, images were captured at 405–490 nm after excitation at 405 nm using an LSM700 microscope. For Alexa Fluor 488 fluorescence, images were captured at 500–630 nm after excitation at 488 nm using an LSM880 or at 494–630 nm after excitation at 488 nm using an LSM700. To observe double-stained samples, Alexa Fluor 488 fluorescence was captured at 500–574 nm after excitation at 488 nm using an LSM880 or at 490–598 nm after excitation at 488 nm using an LSM700. Alexa Fluor 568 fluorescence was captured at 590–712 nm after excitation at 543 nm using an LSM880 or after excitation at 555 nm using an LSM700 equipped with an LP 568 filter. The z-step was set to 500 nm for SAMs. The images were processed using ZEN (Black Edition) software (Carl Zeiss).

Quantification of H3K9me2 and H3K4me1 signals

For double-stained samples, the fluorescence intensity profiles of H3K9me2 and H3K4me1 were measured along a line connecting two perinuclear H3K9me2 signal peaks located on opposite sides of an individual nucleus. Five nuclei from two SAMs were analyzed. For the graph shown in Figure 6d, the y-axis for each nucleus was normalized by setting the maximum fluorescence intensity for each modification to 1. The x-axis for each nucleus was normalized by defining the position of one perinuclear H3K9me2 peak as 0.0 and the other as 1.0. The normalized signal profiles from the five nuclei were averaged. For this calculation, values were interpolated using the approx function in R to perform linear interpolation across a common set of x-values.

AUTHOR CONTRIBUTIONS

YM and HT designed the research, interpreted the data, and wrote the manuscript with input from all authors. YM conducted all the experiments described in this article. RT, AY, and AH performed essential preliminary experiments to develop the immunostaining protocols.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data underlying this article are available in the text and supporting information.

SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article.

Figure S1. Definitions of the vegetative and reproductive SAMs employed in this study. Optical cross-sections of SAMs during the vegetative and reproductive phases are shown. The samples were stained with SR2200 (cell wall) and DAPI (nuclei). The table summarizes the sampling criteria used in this study. PBM, primary branch meristem.

Figure S2. Effect of formaldehyde concentration on H3K9me2 immunostaining in the SAM. (a) Longitudinal sections and 3D reconstruction of the SAM fixed with various concentrations of formaldehyde (FA). The appropriate laser power range for imaging is indicated below each image. Lower FA concentrations enabled the detection of H3K9me2 with reduced laser power. (b) Merged images of H3K4me1 and H3K9me2 (green) and DAPI (blue) staining in the SAM without fixation (0% FA). Although signal intensity was strongest without fixation, both immunostaining signals and DAPI staining were lost in the basal region of the SAM and the leaf primordium. Scale bars, 20 μ m.

Figure S3. Effects of 2-propanol and 1.0% (v/v) Triton X-100 on immunostaining signals. (a, b) Longitudinal section (a) and enlarged view of the boxed region (b) of the SAM treated with 2-propanol. Non-specific fluorescence was detected outside the nuclei at the tip of the SAM. (c, d) Longitudinal view (c) and enlarged image (d) of the SAM permeabilized with 1.0% (v/v) Triton X-100. Perinuclear signals corresponding to H3K9me2 were clearly detected. Scale bars, 20 μ m.

Figure S4. Effect of antigen retrieval using proteinase K on immunostaining signals in the SAM. (a) Longitudinal sections and 3D reconstruction of the SAM after H3K9me2 immunostaining. Different concentrations and incubation temperatures were used for antigen retrieval. Higher concentrations improved signal detection. ProK, Proteinase K; RT, room temperature. (b) Longitudinal sections and close-up view of H3K4me1 immunostaining. Treatment with 0.1% (v/v) Proteinase K caused nuclear shrinkage compared with untreated samples. Scale bars, 20 μ m (a, b left panels) and 5 μ m (b, close-up views).

Figure S5. Effect of trimming basal tissue on signal intensity in the SAM. (a) Diagram illustrating the trimming of residual basal tissue from the SAM. (b) Longitudinal sections of the SAM after immunostaining with and without residual basal tissue. Scale bars, 20 μ m.

Figure S6. Selection of the optimal antibodies for each chromatin modification in the rice SAM subjected to immunostaining. (a) Selection of an anti-H3K4me1 antibody. Left: fluorescence (green); right: merged images with DAPI (blue). Ab8895 (Abcam) produced nuclear signals consistent with known H3K4me1 signals, while MA302A (Takara Bio) did not yield clear nuclear staining. (b) Selection of an anti-H3K4me3 antibody. Ab8580 (Abcam) generated nuclear signals consistent with known H3K4me3 signals, while MA304A (Takara Bio) and 05-1339 (MERCK) showed signals that localized to the nucleolus. (c) Selection of an anti-H3K9me2 antibody. Perinuclear signals indicative of H3K9me2 were detected using ab1220 (Abcam) and MAB10317 (Fujifilm), whereas MA307A (Takara Bio) produced non-specific fluorescence. (d) Selection of

an anti-H3K27me3 antibody. MA323A (Takara Bio) produced nuclear signals consistent with known H3K27me3 signals, while 07-449 (MERCK) did not produce detectable nuclear signals. (e) Selection of an anti-H3S10p antibody. 04-817 (MERCK) and no. 53348 (Cell Signaling Technology) exhibited M phase-specific signals, while ab14955 (Abcam) exhibited signals in the nuclei independent of the cell cycle. Notably, no. 53348 also showed faint signals in the cytoplasm and nucleolus, which were reduced by the addition of NaCl, as shown in Figure S7. The antibodies tested are listed in Table S1. Selected antibodies are indicated in orange. Scale bars, 5 μ m.

Figure S7. Effect of adding NaCl on background signal reduction during H3K9me2 immunostaining. (a, b) Longitudinal section (a) and enlarged view (b) of the SAM stained with a primary antibody solution without NaCl (1% [v/v] BSA). Non-specific fluorescence was observed in the nucleolus and cytoplasm. (c, d) Longitudinal view (c) and enlarged image (d) of the SAM stained with a primary antibody solution containing 500 mM NaCl. Non-specific fluorescence was reduced. Scale bars, 20 μ m (a, b) and 5 μ m (c, d).

Figure S8. Application of iTOMEI clearing for 3D immunostaining of the rice SAM. (a) Orthogonal sections of the rice SAM subjected to 3D immunostaining for H3K9me2. Upper panels: samples mounted in PBS without iTOMEI treatment. Lower panels: samples treated and mounted with iTOMEI. (b) Longitudinal sections of the rice SAM subjected to H3K4me3 immunostaining. Left: image taken immediately after clearing. Middle: same image after 50 days of storage. Right: same image after 140 days of storage. Although higher laser power was required after 140 days, the fluorescence signals were well preserved in samples treated with iTOMEI clearing solution. Scale bars, 20 μ m.

Figure S9. Application of 3D immunostaining to various rice organs and tissues. H3K4me1 immunostaining of a developing leaf sheath (a), mature leaf blade (b), and enlarged views (c); root epidermis (d), root meristem (e), axillary meristem (f), and enlarged views (g and h; h shows a signal-enhanced version of g to facilitate localization); and a developing spikelet (i). For each sample, SR2200 + DAPI (a-1 to i-1), Alexa Fluor 488 (a-2 to i-2), and merged images (a-3 to i-3) are shown. P1-3, First to third youngest leaf primordia. Scale bars, 20 μ m (a, c, e, g, and h), 50 μ m (b and d), and 100 μ m (f and i).

Table S1. Antibodies used in this study.

Data S1. Supplementary text.

Movie S1. Sampling procedure for the vegetative rice SAM.

Movie S2. Sampling procedure for the reproductive rice SAM.

Movie S3. H3K4me1 3D immunostaining of the rice SAM.

Movie S4. H3K4me3 3D immunostaining of the rice SAM.

Movie S5. H3K27me3 3D immunostaining of the rice SAM.

Movie S6. H3K9me2 3D immunostaining of the rice SAM.

Movie S7. H3K9me2 3D immunostaining combined with cell wall staining of the rice leaf primordium.

Movie S8. H3S10p 3D immunostaining of the rice SAM.

Movie S9. H3K4me1 3D immunostaining of the developing rice leaf sheath.

Movie S10. H3S10p 3D immunostaining of the rice axillary meristem.

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